

3 Yr. Degree/4 Yr. Honours 2nd Semester Examination, 2025 (CCFUP)**Subject : Chemistry****Course: CHEM2011****(Basic Chemistry-II)****Time: 2 Hours****Full Marks: 40***The figures in the right hand margin indicate full marks.**Candidates are required to give their answers in their own words
as far as practicable.***1. Answer any five questions:****2×5=10**

- (a) What are ambident nucleophiles? Give one example of such nucleophile showing its nucleophilic centres.
- (b) Identify X and Y, where $(\delta P/\delta T)_V = (\delta S/\delta X)_Y$.
- (c) KHF_2 is a stable compound, but KHBr_2 does not exist. —Why?
- (d) Write down the conditions of spontaneity in terms of ΔG and ΔS .
- (e) What is oxidation number? Give an example of a compound where the constituent element exhibits fractional oxidation number.
- (f) With suitable examples show two tautomerisms other than keto-enol system.
- (g) The van der Waals constants for CO_2 are estimated to be $3.59 \text{ L}^2 \text{ atm mol}^{-2}$ and 43 cc mol^{-1} . What is the inversion temperature of the gas?
- (h) Draw the Fischer Projection of (2R,3S)-3-chloro-2-pentanol.

2. Answer any two questions:**5×2=10**

- (a) (i) When a solution of optically active s-butyl phenyl ketone in aqueous ethanol is treated with acid and base, the solution loses its optical activity. —Explain.
- (ii) What do you mean by “optical purity of an enantiomeric mixture is 63% with respect to 1-isomer”? 3+2
- (b) (i) Which one between H_2O and H_2S molecules has a greater bond angle and why?
- (ii) Between NH_3 and NF_3 , which one is more polar and why?
- (iii) Give an example of a disproportionation reaction. 2+2+1

- (c) (i) A spherical drop of H_2O of 0.4 cm in radius is split up into 125 tiny droplets. Find the increase in the surface energy ($\gamma_{\text{H}_2\text{O}} = 72 \text{ dynes cm}^{-1}$).
 (ii) Explain the variation of viscosity of liquid with temperature.
- (d) (i) Show that surface tension and surface energy have same dimension.
 (ii) The melting point of CaI_2 (575°C) is much lower than that of CaF_2 (1392°C). — Explain
 (iii) How can you differentiate an asymmetric molecule from dissymmetric molecule?

3+2

3. Answer any two questions:

2+2+1

10×2=20

- (a) (i) If the speed distribution function is given by $\frac{dn_c}{n} = A c^m e^{-bc^2} dc$, find dimensions of A and B . Find also the relation between A and B from normalization condition (for $m = 2$).

- (ii) Calculate entropy change for the following process:

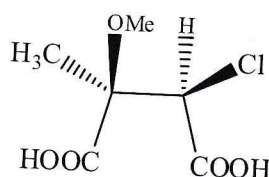
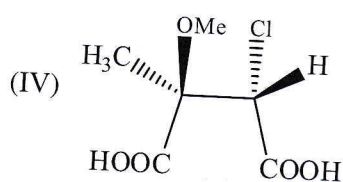
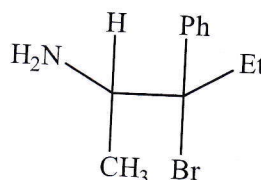
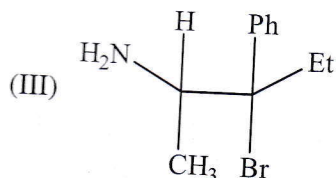
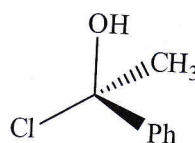
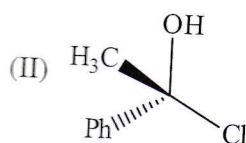
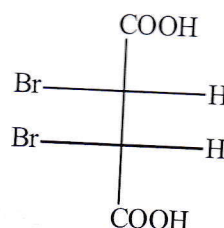
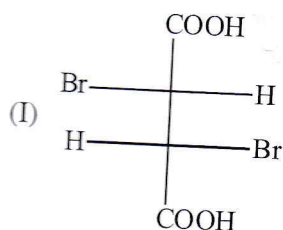
One mole H_2O ($l, -10^\circ\text{C}, 1 \text{ atm}$) $\rightarrow$ One mole H_2O ($s, -10^\circ\text{C}, 1 \text{ atm}$)

Given, $C_p (\text{H}_2\text{O}, l) = 75.42 \text{ J.K}^{-1}$ and $C_p (\text{H}_2\text{O}, s) = 37.20 \text{ J.K}^{-1}$

- (iii) How does Carnot's cycle look like in T-S diagram?

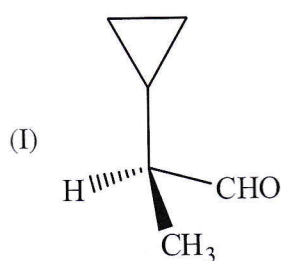
(2+2)+4+2

- (b) (i) Assign the following pairs as enantiomers, diastereomers and homomers.

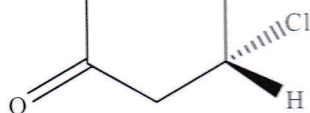


- (ii) Triethylmethylchloride undergoes nucleophilic substitution reaction 10^6 times faster than 1-chlorobicyclo-[2,2,2] octane. —Justify.
- (iii) What is primary kinetic isotope effect? Explain with an example.
- (iv) Which tautomer is favored in 1,3-dicarbonyl compounds? 4+3+2+1
- (c) (i) Balance the following chemical reaction by ion electron method:

$$\text{K}_2\text{Cr}_2\text{O}_7 + \text{H}_2\text{SO}_4 + \text{H}_2\text{S} \rightarrow \text{Cr}_2(\text{SO}_4)_3 + \text{S} + \text{K}_2\text{SO}_4 + \text{H}_2\text{O}$$
- (ii) Which one should have higher bond angle and why?
 F_2O and Cl_2O
- (iii) The dissociation of H_2S decreases in presence of dil. HCl . —Explain.
- (iv) Copper displaces zinc from a solution of zinc salt containing excess KCN . —Explain. 2+3+2+3
- (d) (i) Apply VSEPR theory to predict and draw structure of the following.
 ClF_3 and XeF_4
- (ii) The molecules of a gas are confined to move in a place. Derive the expression for rms speed of the gas.
- (iii) Draw all possible stereoisomers of $\text{Ph CH(OH)CH = CHMe}$.
- (iv) Assign *R/S* designation of the following compounds: 3+2+2+3



(II)



(III)

